Math 161
Problem set 8 solutions

Justin Campbell

1. Since $|f(0)| \leq 0^2 = 0$, we have $f(0) = 0$. Thus
$$f'(0) = \lim_{h \to 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \to 0} \frac{f(h)}{h}.$$
For any $h \neq 0$, we have
$$\left| \frac{f(h)}{h} \right| = \frac{|f(h)|}{|h|} \leq \frac{h^2}{|h|} = \frac{|h|^2}{|h|} = |h|,$$
i.e. $-h \leq |\frac{f(h)}{h}| \leq h$. Since
$$\lim_{h \to 0} -h = 0 = \lim_{h \to 0} h,$$
the squeeze theorem implies that
$$f'(0) = \lim_{h \to 0} \frac{f(h)}{h} = 0.$$

2. If a is a double root of f, then clearly a is a root of f, since
$$f(a) = (a-a)^2 g(x) = 0.$$
As for f', we have
$$f'(x) = 2(x-a)g(x) + (x-a)^2 g'(x),$$
and hence
$$f'(a) = 2(a-a)g(a) + (a-a)^2 g'(a) = 0.$$
Now assume that a is a root of both f and f'. Since a is a root of f, we have $f(x) = (x-a)g(x)$ for some polynomial g by the hint. Thus
$$f'(x) = g(x) + (x-a)g'(x),$$
which implies that
$$0 = f'(a) = g(a) + (a-a)g'(a) = g(a),$$
i.e. a is a root of g. Applying the hint again, we obtain $g(x) = (x-a)h(x)$ for some polynomial h, and hence
$$f(x) = (x-a)g(x) = (x-a)^2 h(x).$$
So a is a double root of f.

3. Because $f(a) = g(a) = 0$, we have
$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \to 0} \frac{f(a+h)}{h},$$
and similarly for g. Note that since
$$\lim_{h \to 0} \frac{g(a+h)}{h} = g'(a) \neq 0,$$

we have $g(a+h) \neq 0$ for h sufficiently close to 0. Thus we obtain

$$\frac{f'(a)}{g'(a)} = \frac{\lim_{h\to 0} \frac{f(a+h)}{h}}{\lim_{h\to 0} \frac{g(a+h)}{h}} = \lim_{h\to 0} \frac{\frac{f(a+h)}{h}}{\frac{g(a+h)}{h}} = \lim_{h\to 0} \frac{f(a+h)}{g(a+h)} = \lim_{x\to a} \frac{f(x)}{g(x)}$$

by the theorem on quotients of limits (in particular the quotients inside the limits are well-defined for h sufficiently close to 0, or equivalently x sufficiently close to a).

4. Define $f : \mathbb{R} \to \mathbb{R}$ by

$$f(x) := \begin{cases} x^{n+1} & \text{for } x \geq 0; \\ -x^{n+1} & \text{for } x < 0. \end{cases}$$

For $x > 0$, we have

$$f'(x) = (n+1)x^n,\ f''(x) = (n+1)nx^{n-1}, \cdots,\ f^{(n-1)}(x) = \frac{(n+1)!}{2}x^2,\ f^{(n)}(x) = (n+1)!x.$$

Similarly, for $x < 0$ we have

$$f^{(n-1)}(x) = -\frac{(n+1)!}{2}x^2 \text{ and } f^{(n)}(x) = -(n+1)!x.$$

Next, we claim that $f^{(n)}(0) = 0$. For this, observe that

$$\lim_{h\to 0^+} \frac{f^{(n-1)}(0+h) - f^{(n-1)}(0)}{h} = \lim_{h\to 0^+} \frac{\frac{(n+1)!}{2}\cdot(0+h)^2 - \frac{(n+1)!}{2}\cdot 0^2}{h}.$$

But we also have

$$\left.\frac{\mathrm{d}}{\mathrm{d}x}\right|_{x=0} \frac{(n+1)!}{2}x^2 = \lim_{h\to 0} \frac{\frac{(n+1)!}{2}\cdot(0+h)^2 - \frac{(n+1)!}{2}\cdot 0^2}{h} = \lim_{h\to 0^+} \frac{\frac{(n+1)!}{2}\cdot(0+h)^2 - \frac{(n+1)!}{2}\cdot 0^2}{h},$$

and

$$\left.\frac{\mathrm{d}}{\mathrm{d}x}\right|_{x=0} \frac{(n+1)!}{2}x^2 = (n+1)!\cdot 0 = 0.$$

An analogous argument shows that

$$\lim_{h\to 0^-} \frac{f^{(n-1)}(0+h) - f^{(n-1)}(0)}{h} = \left.\frac{\mathrm{d}}{\mathrm{d}x}\right|_{x=0} - \frac{(n+1)!}{2}x^2 = -(n+1)!\cdot 0 = 0.$$

Thus we have shown that

$$f^{(n)}(0) = \lim_{h\to 0} \frac{f^{(n-1)}(0+h) - f^{(n-1)}(0)}{h} = 0$$

as claimed.

Assembling the above information, we obtain that f is n times differentiable, and that

$$f^{(n)}(x) = (n+1)!|x|$$

for all $x \in \mathbb{R}$. This formula implies that $f^{(n)}(x)$ is not differentiable at 0, and hence f is not differentiable $n+1$ times at 0.

5. (a) Using the various differentiation rules, we see that

$$f'(x) = 2x\sin\tfrac{1}{x} + x^2(-\tfrac{1}{x^2}\cos\tfrac{1}{x}) = 2x\sin\tfrac{1}{x} - \cos\tfrac{1}{x}$$

for $x \neq 0$. Since $|\sin\frac{1}{x}| \leq 1$ for all $x \neq 0$, we have

$$|f(x)| = x^2|\sin\tfrac{1}{x}| \leq x^2$$

for all $x \neq 0$. Combining this with $f(0) = 0$, it follows that $|f(x)| \leq x^2$ for all $x \in \mathbb{R}$. Thus, by problem 1 we have $f'(0) = 0$ (this was not part of the statement, but was established in my solution). So f is differentiable everywhere as claimed.

(b) We claim that $\lim_{x\to 0} f'(x)$ does not exist, from which it follows that f' is not continuous at 0. As was shown in class, the squeeze theorem implies that

$$\lim_{x\to 0} x \sin \tfrac{1}{x} = 0.$$

Thus, if $\lim_{x\to 0} f'(x)$ exists, then by the formula for $f'(x)$ (when $x \neq 0$) obtained in (a),

$$\lim_{x\to 0} \cos \tfrac{1}{x} = \lim_{x\to 0} (2x \sin \tfrac{1}{x} - f'(x)) = 2 \cdot 0 - \lim_{x\to 0} f'(x) = -\lim_{x\to 0} f'(x)$$

exists. But the limit on the left side does not exist, since $\cos \frac{1}{x}$ oscillates between 1 and -1 as $x \to 0$ (this is the same reason that $\lim_{x\to 0} \sin \frac{1}{x}$ does not exist, which we proved in class).